

EEE 359

Engineering Project Management

Lecture 08

Scheduling Methods

Gantt Charts

Gantt Chart

- A Gantt chart is a **bar chart** that's widely used for scheduling and tracking tasks over time.
- Uses a **horizontal time scale** (days, weeks, months) to show when tasks start and finish.
- Each activity is represented by a horizontal bar:
 - The **position** shows the start date.
 - The **length** shows the duration.
- Tasks are arranged in order, often showing dependencies (FS, SS, FF, SF).

Project Planner

Select a period to highlight at right. A legend describing the charting follows.

Period Highlight: 1

Plan Duration

Actual Start

■ % Complete

Actual (beyond plan)

 % Complete (beyond plan

[illegible]

Europa project

VanArsdel, Ltd. Project lead

SIMPLE GANTT CHART by Vertex42.com

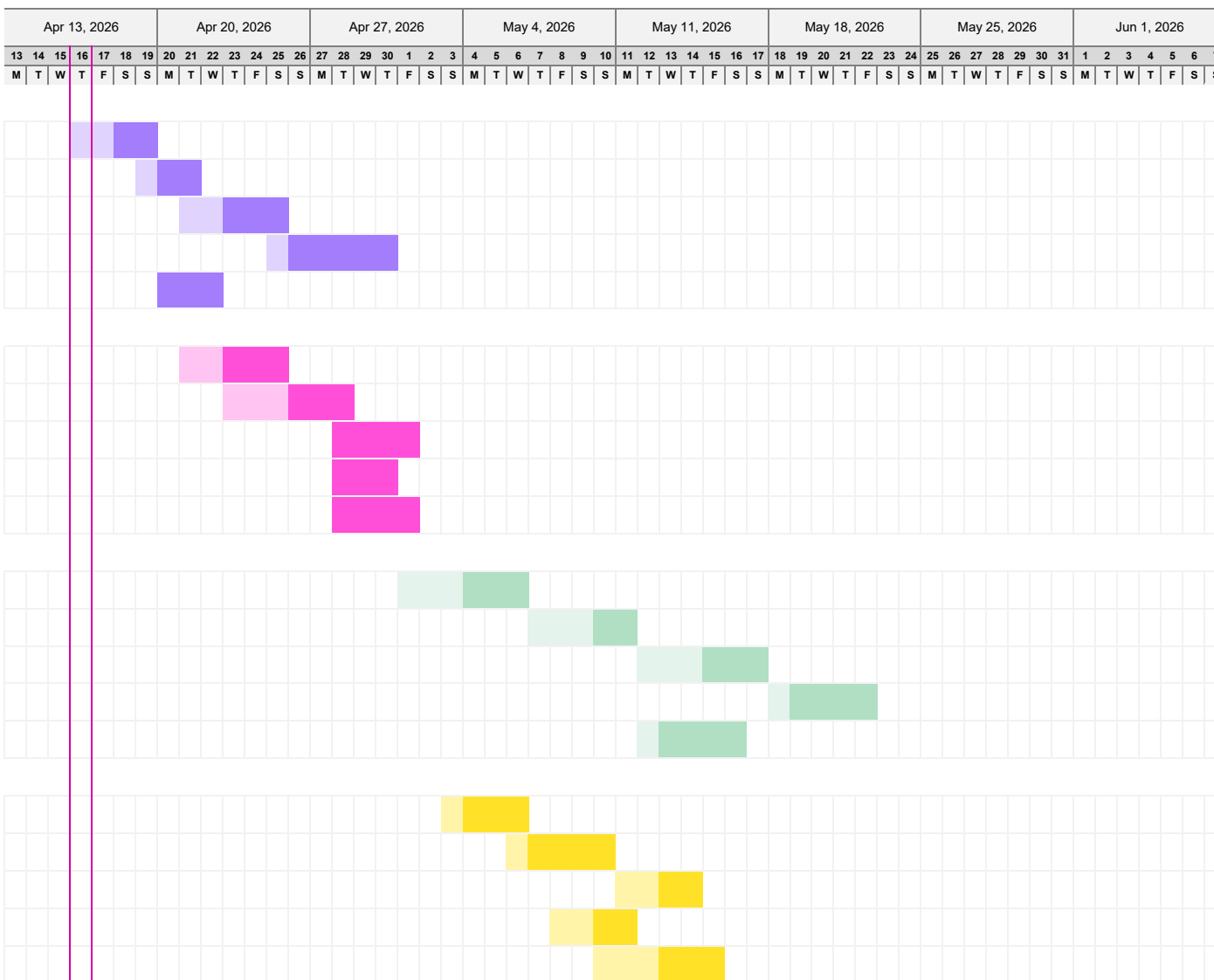
<https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html>

Task	Assigned to	Progress	Start	End
Initiation				
Define goals	Gokce Aslan	50%	4/16/26	4/19/26
Conduct studies	Hayden Cook	60%	4/19/26	4/21/26
Establish comms	Jens Martensson	50%	4/21/26	4/25/26
Develop charter	Nuria Acevedo	25%	4/25/26	4/30/26
Set up team	Olivia Wilson		4/20/26	4/22/26
Planning and design				
Create schedule	Gokce Aslan	50%	4/21/26	4/25/26
Identify deliverables	Hayden Cook	50%	4/23/26	4/28/26
Develop budget	Jens Martensson		4/28/26	5/1/26
Define scope	Nuria Acevedo		4/28/26	4/30/26
Identify risks	Olivia Wilson		4/28/26	5/1/26
Execution				
Execute tasks	Gokce Aslan	50%	5/1/26	5/6/26
Monitor progress	Hayden Cook	60%	5/7/26	5/11/26
Manage resources	Jens Martensson	50%	5/12/26	5/17/26
Provide updates	Nuria Acevedo	25%	5/18/26	5/22/26
Testing and validation	Olivia Wilson	25%	5/12/26	5/16/26
Evaluation				
Monitor progress	Gokce Aslan	25%	5/3/26	5/6/26
Track expenses	Hayden Cook	25%	5/6/26	5/10/26
Evaluate progress	Jens Martensson	50%	5/11/26	5/14/26
Address risks	Nuria Acevedo	60%	5/8/26	5/11/26
Gather feedback	Olivia Wilson	50%	5/10/26	5/15/26

Insert new rows ABOVE this one

Project start: Thu, 4/16/2026

Display week: 1



Milestone

Important deadlines in a project/program are highlighted by specific points in time called **milestones**. These are timeless activities usually at the **beginning or end of a phase or stage** and are used for **monitoring purposes** throughout the life of the project. Needless to say, they should be **SMART**, which is an acronym for Specific, Measurable, Achievable, Realistic, Timebound.

Often milestones are used to act as **trigger points** for progress payments or deadlines for receipt of vital information, permits or equipment deliveries.

Milestone reports are a succinct way of **advising top management of the status** of the project and should act as a spur to the project team to meet these **important deadlines**. This is especially important if they relate to large amount of progress payments.

Milestones are marked on **bar charts or networks by a triangle or diamond** and can be turned into a monitoring system in their own right when used in milestone slip charts, sometimes also known as trend charts.

Critical Path

Critical Path Method

- Critical path is the sequence of activities that represents the longest path through a project, which determines the shortest possible project duration.
- Estimates the minimum project duration and determines the amount of scheduling flexibility on the logical network paths within the schedule module.
- A critical path is normally characterized by **zero** total float on the critical path.

Near Critical Path

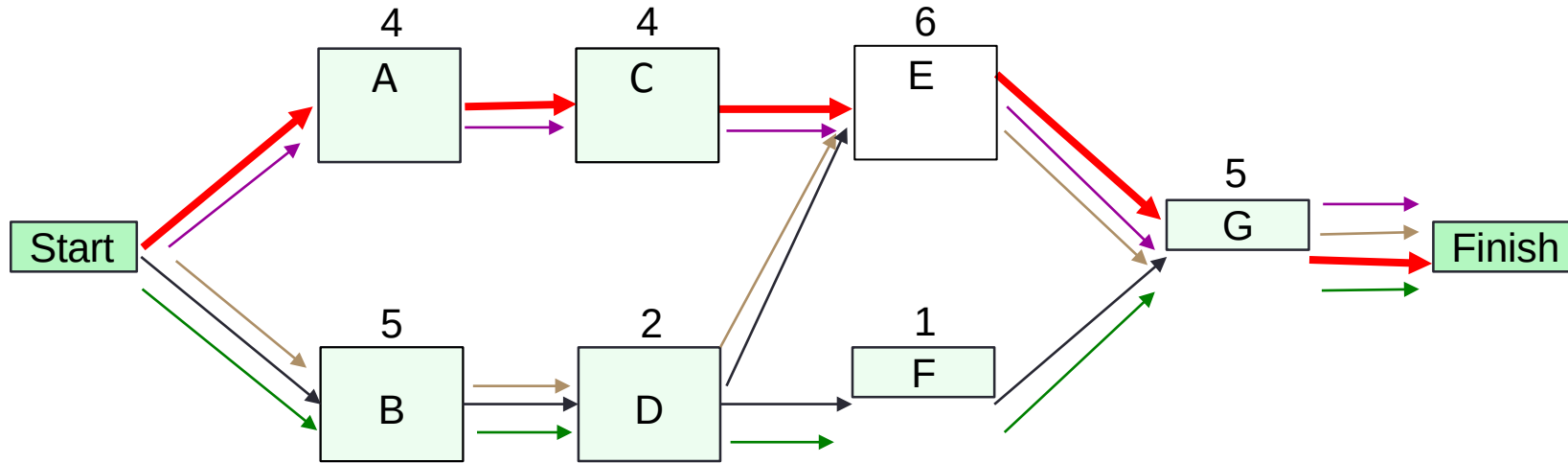
- Path(s) closest in length to the critical path
- The near-critical paths can become critical
- The closer in length the near-critical and critical paths are, the more risk to the project
- The project manager needs to focus both time and effort monitoring and controlling critical and near-critical path activities, so as not to delay project completion

Critical Path Calculation

When the project starts, Activities A and B can begin. When Activity A is done, Activity C can begin. When Activity B is done, Activity D can begin. Activities C and D must finish before Activity E can begin. Activity F can begin when Activity D is completed. Activity G can begin when Activities E and F are completed. When Activity G is completed, the project is complete. The durations are shown in the table.

Activity	Duration
A	4 Days
B	5 Days
C	4 Days
D	2 Days
E	6 Days
F	1 Days
G	5 Days

Critical Path Calculation



- What are the different paths and times?

- Path A,C,E,G Duration $4+4+6+5 = 19$
- Path B,D,E,G Duration $5+2+6+5 = 18$
- Path B,D,F,G Duration $5+2+1+5 = 13$

- What is the critical path?

- Path A,C,E,G Duration of 19

Critical Path

More complicated network diagrams use the **Forward Pass- Backward Pass** method to calculate:

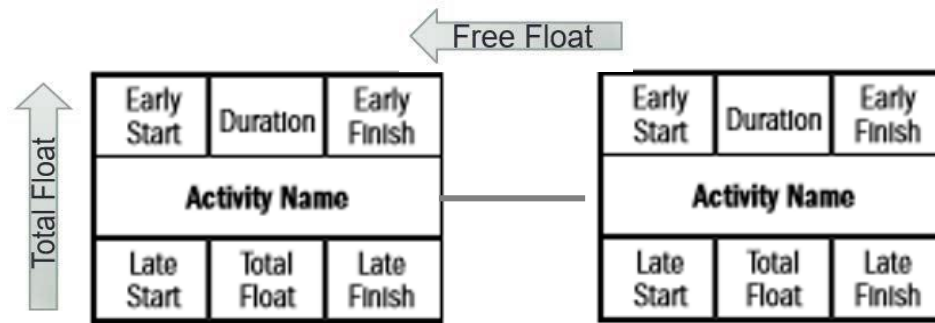
- Total Float /Slack.
- Early start/finish dates
- Late start/finish dates.

Early Start	Duration	Early Finish
Activity Name		
Late Start	Total Float	Late Finish

These dates are **possible dates** *for* the project schedule.

- They indicate the time periods within which the activity could be executed.
- Also use the data entered in the schedule model for activity durations, logical relationships, leads, lags, and other known constraints.

Float

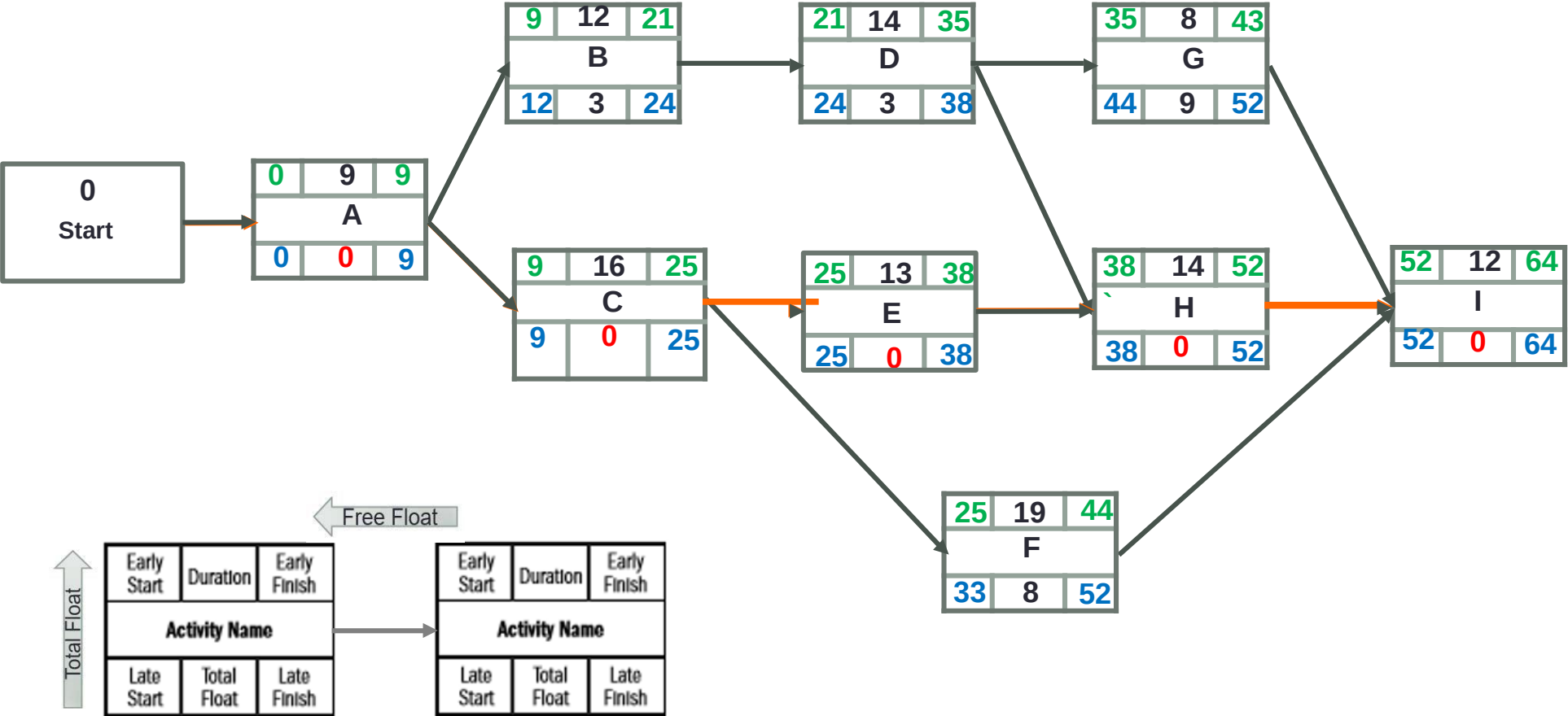


- **Total float: (TF)** = $LS - ES$ or $LF - EF$ the amount of time an activity can be delayed without delaying **the project end date**
- **Free float: (FF)** = $ES \text{ (successor)} - EF \text{ (present activity)}$ the amount of time an activity can be delayed without delaying the **early start date of its successor(s)**.
- **Zero float:** the task that has no extra amount of time. These tasks sit on the critical path, and any delay here pushes back the entire project.

Float Calculation

Activity	Predecessor	Duration (days)
A	--	9
B	A	12
C	A	16
D	B	14
E	C	13
F	C	19
G	D	8
H	D,E	14
I	F,G,H	12

Float Calculation



Thanks